Factsheet
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CONCEPT AMG GT XX – a New Dimension of Performance

Did you know that…

- The CONCEPT AMG GT XX features three revolutionary axial flux motors that together produce more than 1,000 kW (1,360 hp). That's despite each one being around two-thirds smaller and lighter than a traditional electric motor. In fact, they are so compact that each motor would almost fit inside a pizza box.

- The electric motors are packaged into High Performance Electric Drive Units (HP.EDUs) front and rear, with one motor in the front drive unit and two in the rear. Each motor assembly features its own silicon-carbide DC/AC inverter. One of the hardest substances on earth – with an official rating just below diamonds – silicon carbide is extremely tough and heat-resistant, which is why it's also used in rocket engines and as protective armour on space telescopes.

- Copper coils are an integral part of an electric motor, but the key is to pack as much of the material as possible into a very tight space. That's why the copper wire used in the innovative axial flux motor that powers the Concept AMG GT XX isn't round, but rectangular. Think of it like pasta: when serving up a hearty Ragù, Italians swap spaghetti for tagliatelle, because its flatter, broader surface carries more of the sauce. The rectangle-shaped wire is then precisely bent and wound into a specific radius, allowing for more copper per slot in the stator. The result is better power density and thermal efficiency – and not a tomato-stained shirt in sight.

- Production of the axial flux motor alone involves around 100 production processes, 65 of which are new to Mercedes-Benz and 35 of which are world firsts. This staggering number includes innovative bonding processes and groundbreaking use of artificial intelligence – as well as some serious attention to detail. That's because to connect the stator, the surface has to be 100 per cent dust- and grease-free, so an AI-supported vision system is used, combining powerful sensors and cameras, to place 'virtual protection zones' over sensitive areas. It's a little bit like masking your switches and door frames before painting your walls, but with a level of accuracy that's impossible for the human brain to handle. A laser is then used to clean the material with pinpoint accuracy.

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- The CONCEPT AMG GT XX has a drag coefficient of 0.198 – about the same as a soaring eagle. Very few things in nature have a more aerodynamic shape – except for a peregrine falcon in full dive mode or a dolphin at full speed. This slipstreamed design increases driving range while also maximising handling stability in all areas up to a top speed of more than 360 km/h, without the need for active aero elements.

- It's not just the headlights that shine – sections of the car's bodywork also light up with the MBUX Fluid Light Paint. This electroluminescent technology works by applying a small current to electrically conductive paint layers. As well as producing a cool visual effect at night, this electric paint also signals when the car is slowing to a standstill, when it's ready to charge and how far charging has progressed. Such paint's properties also make it well suited to covert operations and it has even been used for experimental aircraft markings. Useful for slipping under the radar…

- In a patented world first, the CONCEPT AMG GT XX features wheels with active aero elements, providing the perfect answer to the conflict between brake cooling and aerodynamic efficiency. Each hub assembly has its own self-sufficient system comprising a generator, actuator and battery. When actuated, the wheel's five aeroblades create an opening around the edge of the rim and cooling air is pumped through to the brakes and back out again.

- Like many EVs, the CONCEPT AMG GT XX has external speakers to emit the driving sound and for pedestrian protection, but for the first time ever they are integrated into the headlight housings. These innovative speakers save space, reduce weight and enable new sound effects. They also feature passive membrane tech, which uses air pressure to produce deep bass and low frequency sounds without any electrical input.

- For the first time in the automotive world, the CONCEPT AMG GT XX uses an innovative material called LABFIBER Biotech Leather Alternative. Produced with the latest biotechnology, it has the same look and feel as genuine leather yet it's twice as strong and, with low thermal conductivity, it doesn't heat up as quickly – even in summer. Its clever combination of materials includes vegetable proteins, bio-based polymers and, best of all, a recycled AMG GT3 racing tyre. So our motorsport DNA doesn't just put you in the driver's seat – it *is* the driver's seat.

- The carbon-fibre front bucket seats feature ergonomic pads created using a 3D printing process and are tailored specifically to each driver. The idea comes from the racing world, where individual seat pads can be quickly swapped out for driver changes in endurance races. Every pad is finished with LABFIBER Biotech Leather Alternative, which includes material from a recycled AMG GT3 tyre, meaning motorsport is literally part of the fabric of the car.

- The car's interior door handles – finished in orange to match the exterior – are made from a material called LABFIBER Biotech Silk Alternative. Designed to replicate spider's silk, it's both stronger and lighter than traditional materials such as nylon or polyester. It's produced by harvesting silk proteins from genetically modified bacteria with silk-producing superpowers, and processing it into a strong, shiny yarn. It's 100 per cent biodegradable, tuneable and scalable.

- The highlight at the rear is an innovative MBUX Fluid Light Panel with more than 700 programmable LEDs providing a 3D-pixel effect with text and animation. It can display a variety of content, from a digital AMG logo to the current state of charge – handy for checking battery level at a glance while you're having a coffee. When not in use the panel blends into the bodywork, so you'd never know it's a secret screen.

- The CONCEPT AMG GT XX has a very high average charging power of more than 850 kW over a wide range of the charging curve. In just five minutes it can add 400 km of range[1] – enough to drive from Affalterbach to Spa-Francorchamps (with enough range left for a few fast laps). To match that extreme charging performance, we worked closely with Alpitronic (a leader in high-power charging) on a prototype charging station – the first capable of transmitting such a high current via a standard CCS connector.

- It's no coincidence that the CONCEPT AMG GT XX bears visual similarities to the C111 and Vision One-Eleven, two aerodynamic icons of their time that were also swathed in evocative orange and black. In the 1960s and 1970s, the experimental C111 series was used for testing new Wankel and turbo diesel engines. Today, the CONCEPT AMG GT XX pioneers a revolutionary electric performance drive concept: the axial flux motor.

Key technical data at a glance

CONCEPT AMG GT XX		
Electric motors	Type	Three axial flux motors
Max. drivetrain power	kW (hp)	>1,000 (>1,360)
Drive		AMG Performance 4MATIC+ fully variable all-wheel drive
Top speed	km/h	>360
Charging time for approximately 400 km range[1]	min	~5
Average DC charging power over a wide SOC range	kW	>850
Nominal voltage	Volt	>800
Length/width/height	mm	5,204/1,945 (2,130 with mirrors)/1,317
C_d value		0.198
Frontal area	m^2	2.24

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Press information and digital services for journalists and multipliers can also be found on our **Mercedes-Benz Media online platform** at media.mercedes-benz.com.

Mercedes-Benz AG at a glance
Mercedes-Benz AG is part of the Mercedes-Benz Group AG with a total of around 175,000 employees worldwide and is responsible for the global business of Mercedes-Benz Cars and Mercedes-Benz Vans. Ola Källenius is Chairman of the Board of Management of Mercedes-Benz AG. The company focuses on the development, production and sales of passenger cars, vans and vehicle-related services. Furthermore, the company aspires to be the leader in the fields of electric mobility and vehicle software. The product portfolio comprises the Mercedes-Benz brand with Mercedes-AMG, Mercedes-Maybach and G-Class with their all-electric models as well as products of the smart brand. Mercedes-Benz AG is one of the world's largest manufacturers of high-end passenger cars. In 2024 it sold around 2,4 million passenger cars and vans. In its two business segments, Mercedes-Benz AG is continually expanding its worldwide production network with more than 30 production sites on four continents, while gearing itself to meet the requirements of electric mobility. At the same time, the company is constructing and extending its global battery production network on three continents. As sustainability is the guiding principle of the Mercedes-Benz strategy and for the company itself, this means creating lasting value for all stakeholders: for customers, employees, investors, business partners and society as a whole. The basis for this is the sustainable business strategy of the Mercedes-Benz Group. The company thus takes responsibility for the economic, ecological and social effects of its business activities and looks at the entire value chain.

[1] The information is preliminary. The range values are provisional, based on digital simulations of the standard determination for WLTP. So far, there are neither confirmed values from an officially recognized testing organization nor an EC type approval nor a certificate of conformity with official values. Discrepancies between the provided information and the official values are possible. Binding values are not yet available.